Soma: Cost, Deployment and Operations

Nov 3, 2026 · @Varun Prakash



Summary

Soma, one cislunar ring for about 550,000 people, costs roughly **$90–360 billion** to build, once 100 rings share the cost of a lunar industry, and about **$3–11 billion a year** to run. From the first lunar seed factory to the first residents takes about 20 years.

- **What it is:** a 109 km ring with a 150 m tube, spinning once every 4.4 minutes for 1 g, in orbit about 200,000 km from Earth.
- **What it's made of:** about 300 Mt of mostly lunar material; Earth supplies only electronics, biology and specialist parts.
- **How it runs:** about 3.4 GW of solar power from wings on the six spokes, 6 GWh of batteries for Earth's shadow, and closed loops for air, water, food and waste in which plants and the mountain filters do most of the work.
- **How it starts:** power and cooling first, then sealed and pressurised districts, then machines that clean air and water, then crops, then the forests and wetlands that eventually take over most of that work.
- **Biggest uncertainties:** the cost and pace of lunar industry, how much nitrogen the polar ice really holds, and how reliably the biological loops run at this scale. A 12-year, $37–103B proof-of-concept program, ending with a 150-person Pathfinder ring, tests each of them first.

Stacking rings. Six rings can share one 14 km spine. Build cost per resident falls from $168–661k for ring 1 to $87–502k for ring 100, mostly from reusing the construction fleet and cheaper Earth-made parts.

All figures are order-of-magnitude estimates for a concept.

A day on the ring (91 s animation): terrace homes, a Roadster climbing home, the spoke elevator, farms, the waterfront and the ring at night.



Full video:

Image — A day on the ring, full video

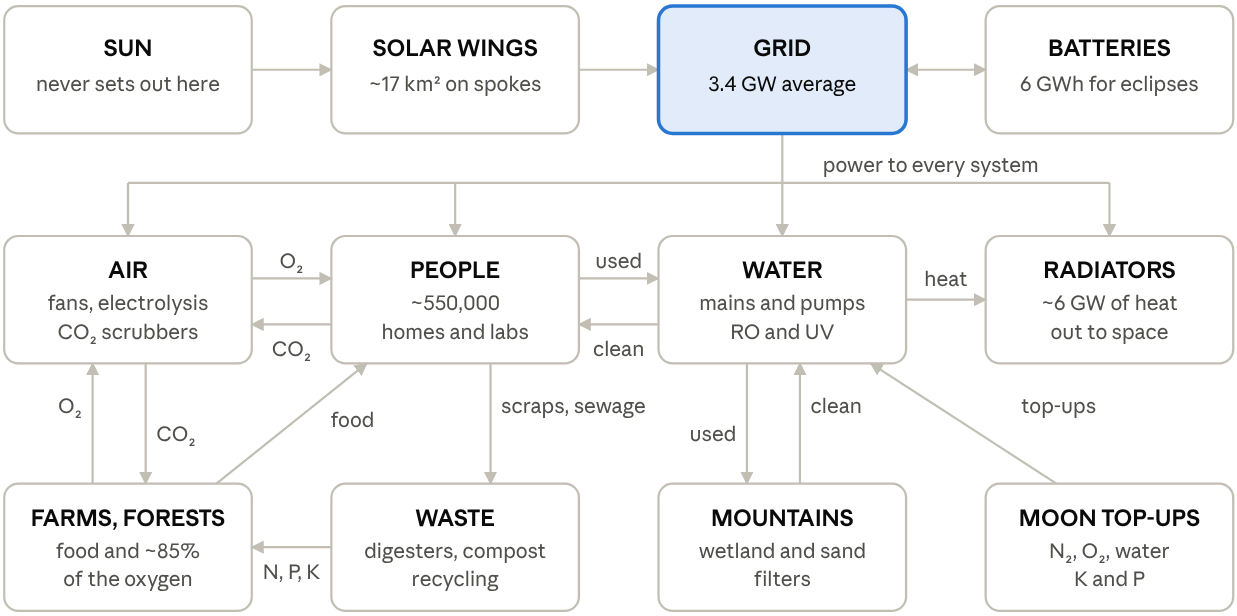
How Soma is built (5 min 44 s film): every construction animation in build order, from the spine to the finished six-ring stack, ending with a zoom out to its orbit around the Moon.

Image — How Soma is built, full film

The ring at a glance

Every system feeds at least two others: power runs everything, plants make most of the oxygen and food, and the mountains clean the water.

Power runs every system; plants tie air, water and food together



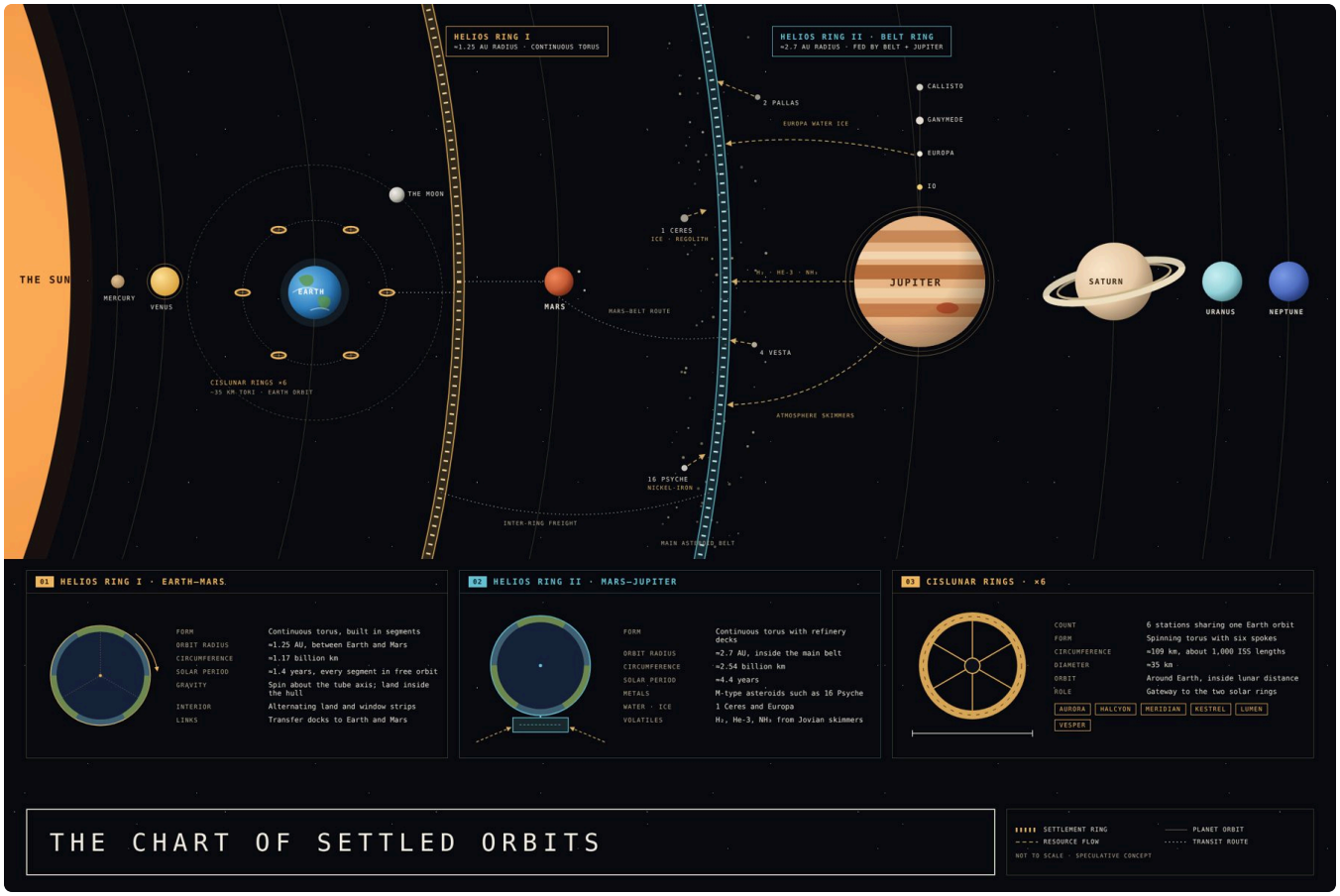
how the systems connect · 12 systems

Power flows down from the wings to every system; air, water, food and waste form loops through people, plants and the mountains.

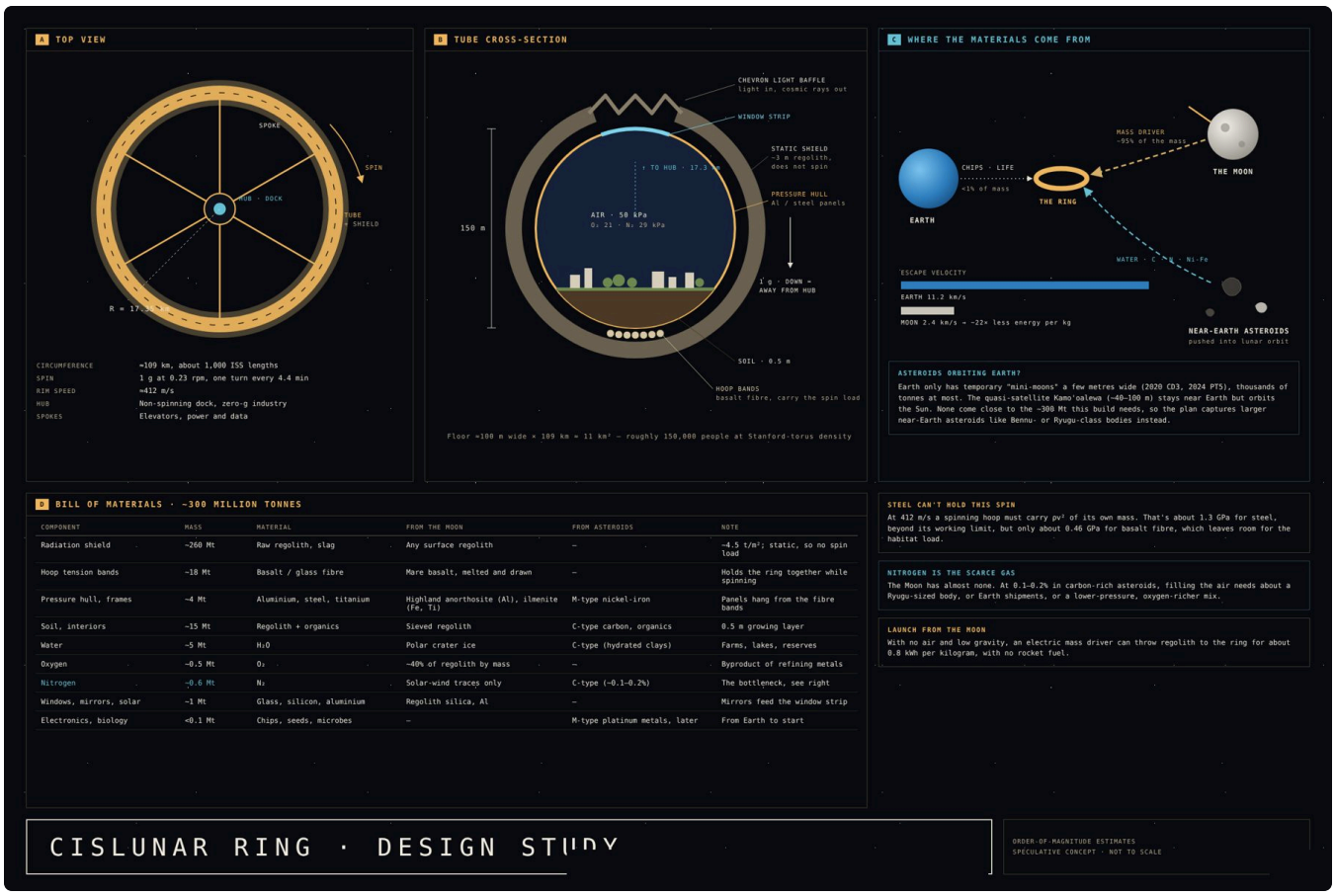
Item	Value
Ring	109 km around, 17.35 km radius, 1 g at 0.23 rpm
Tube	150 m across, 45 levels of 3.33 m, 606 cells of 180 m
Cells	30 districts of 20 cells + 6 spoke junctions; 30% leisure
Population	≈550,000 (stacked homes); ≈375,000 with a terrace-home mix
Atmosphere	57 kPa, 34% O_2, ~0.9 Mt of gas
Water	~5 Mt in tanks, lakes and the river
Power	3.4 GW average, 4.5 GW peak, 81 GWh a day
Solar	~17 km² of wings, ~2.9 km² per spoke
Batteries	~6 GWh in 30 district banks

Item	Value
Radiators	~7.3 km² of fins, one per cell
Orbit	~200,000 km from Earth, 10.3-day period

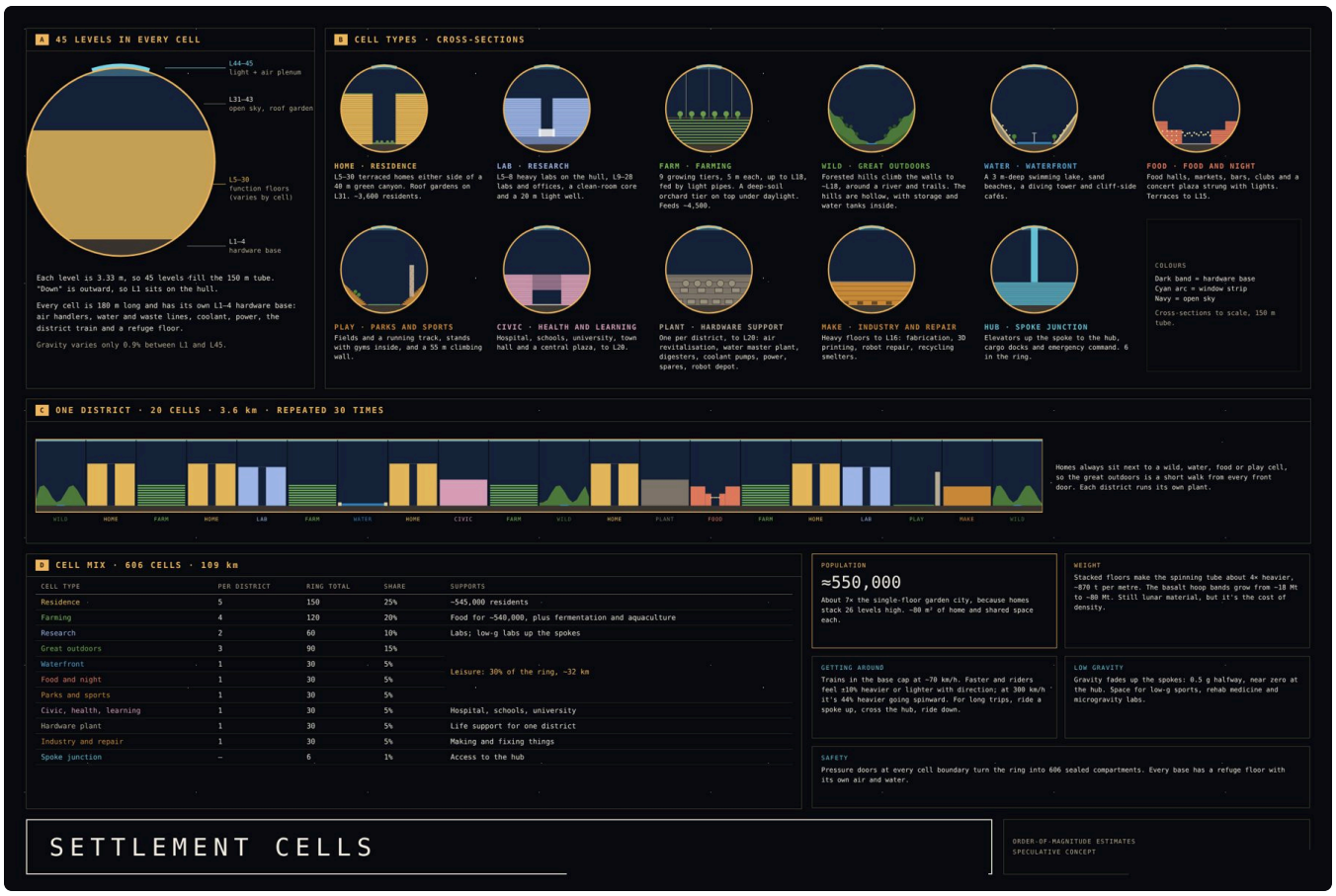
Where it sits. Two Sun-orbiting rings plus the six cislunar rings, three counter-rotating pairs.



The ring's design and materials. Top view, tube cross-section, where the ~300 Mt comes from, and the bill of materials.



The cells. 45 levels per cell, eleven cell types, one 20-cell district and the cell mix.

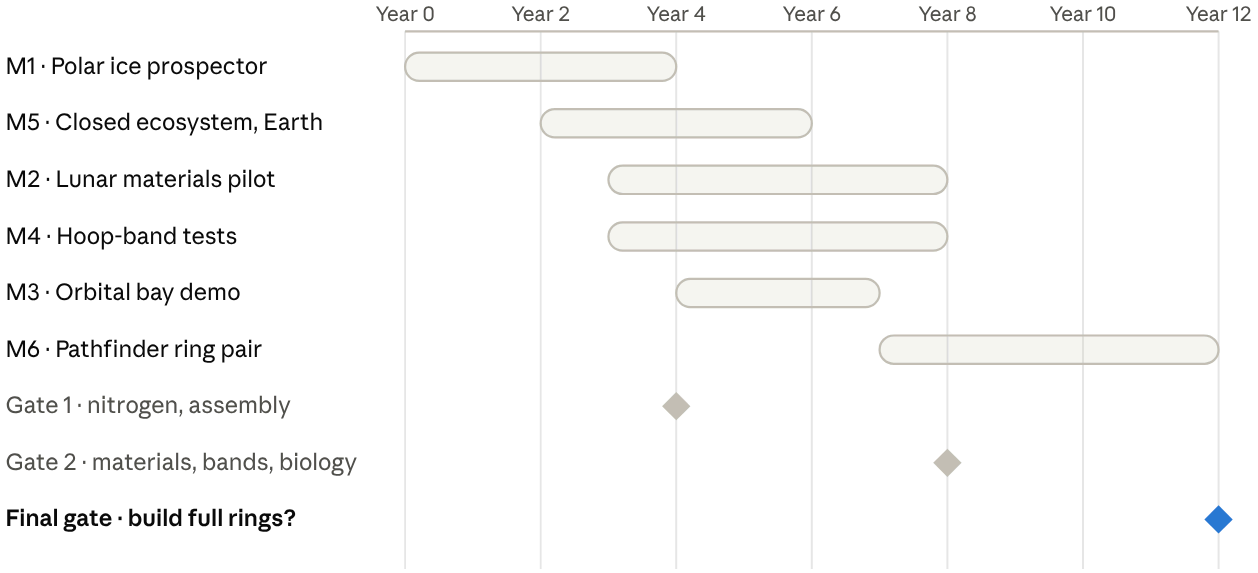


Proof of concept first

Before committing to full rings, six smaller milestones test every major risk over about 12 years for roughly **$37–103 billion**, 10–40% of one ring. The flagship is **Pathfinder**: a small ring pair with 150 people living at 1 g for five years.

Every big risk is tested within 12 years, before a full ring is committed

Proof-of-concept milestones and go/no-go gates, years from the start



proof-of-concept plan · 6 milestones, 3 gates

The cheap tests on Earth and the Moon come first; Pathfinder's build starts after Gate 1, and its crew moves in only after Gate 2.

#	Milestone	Risk it tests	Test	Passes if	Years	Cost
M1	Polar ice prospector	Nitrogen and water supply	Rovers and drills in 3+ cold traps; pilot extraction plant	Ice-rich soil holds ≥0.1% nitrogen across large deposits; 1 t of N_2 and 100 t of water made	0–4	$2–6B
M2	Lunar materials pilot	Lunar industry	Oxygen from molten regolith, aluminium from highland rock, basalt fibre, a 300 m mass driver	10 t/yr of O_2, 100 t/yr of Al, fibre ≥1 GPa, 1 kg packets thrown at 2.4 km/s and caught in orbit	3–8	$10–25B
M3	Orbital bay demo	Robotic assembly, leak rate	A bay-factory prototype builds 3 full-size 18 m bays from Starship-delivered panels	One panel placed an hour; seams leak less than the 0.1%/yr budget; 3 bays in 6 months	4–7	$3–7B

#	Milestone	Risk it tests	Test	Passes if	Years	Cost
M4	Hoop-band qualification	Fibre bands carrying the spin	Full-size band sections on a ground spin rig; 5-year creep tests, incl. lunar-made fibre from M2	Holds 1.5× design load; creeps <0.5% in 5 years	3–8	$1–2B
M5	Closed ecosystem on Earth	Biology at scale	A sealed 2 ha habitat with farm tiers and a mountain filter, 40 people for 2 years	⩾95% of air, water and food recycled; the mountain filter meets drinking standard before the final RO	2–6	$1–3B
M6	Pathfinder ring pair	Spin-up, counter-rotation, Sun-pointing, spoke wings, life support in space, health	224 m ring at 2 rpm plus a counter-rotating ballast ring, 150 crew for 5 years	Crew health stable at 1 g for ⩾2 years; pair spins up and tracks the Sun; ⩾95% closure in space	7–12	$20–60B
	Total				**0–12**	**≈ $37–103B**

M1 and M2 are also the start of the lunar program, so about a third of this would be spent anyway.

Pathfinder in numbers.

Item	Value
Ring	224 m radius, 1.4 km around, 2 rpm for 1 g
Tube	20 m across; ~20,000 m² of floor
Rim speed	~47 m/s; aluminium carries the spin easily, so no fibre bands needed yet
Crew	150, with ~6,800 m² of crops on 3 tiers
Partner	A counter-rotating ballast ring of ~20 kt of bagged lunar regolith, testing the pair principle
Shield	~480 kt of lunar regolith from the M2 mass driver, testing the supply chain too

Item	Value
Power	~2 MW from ~7,000 m² of spoke wings; ~4 MWh of batteries
From Earth	~25 kt: hull panels, life support, robots; ~1,750 launches incl. tankers
Extra test	Spin can be slowed for months at 0.38 g (Mars) or 0.16 g (Moon) to study partial gravity

Pathfinder's cost: ~25 kt delivered at $100–700/kg is $2.5–17.5B; space-rated hardware $10–30B; five years of operations $5–15B.

$$\omega = \sqrt{g / r} = \sqrt{9.81 / 224} \approx 0.209\ \text{rad/s} \approx 2\ \text{rpm}$$

Go/no-go gate at year 12. Full rings start only if every box is ticked:

- [ ] Polar nitrogen is proven, or a cheaper nitrogen plan exists
- [ ] Lunar plants make fibre, aluminium and oxygen at pilot rates, and the mass driver delivers to orbit
- [ ] Robots build at least one bay a week per factory, within the leak budget
- [ ] Fibre bands pass the load and creep tests
- [ ] Closed ecosystems hold ≥95% recycling for 2 years, on Earth and on Pathfinder
- [ ] Pathfinder's crew stays healthy at 1 g for ≥2 years, and the pair spins up and tracks the Sun
- [ ] Delivered cost per kg is within 2× of the plan

Cost to build

A ring costs about **$89 billion** (optimistic) to **$356 billion** (conservative) when 100 rings share the lunar industry. Built alone, the first ring would carry the whole lunar program and cost $0.6–2.2 trillion.

Getting mass to where it's needed. Each Starship delivery beyond low orbit needs tanker flights to refuel it first:

$$c_{\text{delivered}} = \frac{(1 + N_{\text{tankers}})\, c_{\text{launch}}}{m_{\text{payload}}}$$

Destination	Tankers per delivery	At $1.5M a launch	At $10M a launch
Ring orbit, 100 t	~6	~$100/kg	~$700/kg
Lunar surface, 100 t	~10–15	~$250/kg	~$1,600/kg

The lunar industry, shared by 100 rings. Earth lands a seed factory, the Moon builds most of its own expansion, and Earth keeps supplying the precise 10%.

Lunar program item	Optimistic	Conservative
Seed factory, 20,000 t built and landed	$10B	$40B
Earth-made parts for full build-out, ~250,000 t	$135B	$625B
R&D and program	$100B	$200B
Spares and operations, 50 years	$250B	$1,000B
Total	**≈ $0.5T**	**≈ $1.9T**

One ring.

Item	Basis	Optimistic	Conservative
Share of lunar industry	program ÷ 100 rings	$5B	$19B
Earth high-tech kit	~50,000 t of electronics, life-support machines, lab gear, biology	$30B	$60B
Nitrogen	0.3–0.6 Mt from Earth; near zero if polar ice delivers	$30B	$200B
Power system	6 GWh of batteries (~37,500 t from Earth), inverters, grid	$9B	$42B
Robots, bay factories, assembly	6 factories × 7 flights, robot fleet	$5B	$15B
Design, integration, start-up	to the first residents	$10B	$20B
Total		**≈ $89B**	**≈ $356B**

Batteries: 6 GWh × $100/kWh = $0.6B of cells, plus 37,500 t delivered at $100–700/kg = $3.8–26B. Solar wings, radiators, hull and shield come from lunar material, so they sit inside the lunar-industry line.

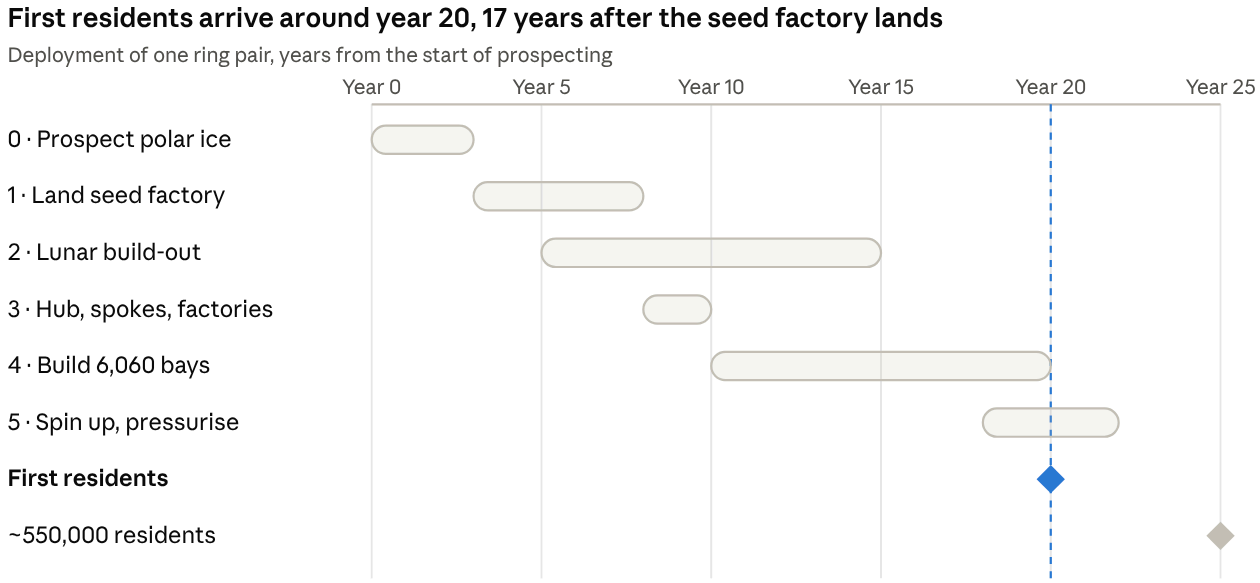
Per home. With 2.5 people per household there are ~220,000 homes. If homes carry 60% of the cost and financing adds 40%:

$$P_{\text{home}} = \frac{C_{\text{ring}} \times 0.6 \times 1.4}{220{,}000} \approx \$340\text{k} \text{ to } \$1.36\text{M}$$

A terrace-home mix (~150,000 homes) raises that to about $0.5–2.0M.

Deployment

The first residents move in around year 20 and the ring is full by about year 25; the pace is set by the lunar mass driver, not by Starship.



deployment plan · years 0–25

Lunar work overlaps the ring build: the Moon keeps expanding while the first bays go up.

Phase	Years	What happens	Starship launches (incl. tankers)
0 · Prospect	0–3	Map polar ice and its ammonia; pilot oxygen and metal plants	~130

Phase	Years	What happens	Starship launches (incl. tankers)
1 · Seed factory	3–8	Land 20,000 t: mining robots, refinery, fibre plant, solar-cell line, first mass driver	~2,600
2 · Lunar build-out	5–15	The Moon builds most of its own expansion; Earth sends ~50,000 t of precise parts in this decade	~6,500
3 · Ring-orbit base	8–10	Hub, spoke cores, mass-driver catcher, 6 bay factories	~640
4 · Build the ring	10–20	6,060 bays, shield, hoop bands, solar wings, radiators; Earth sends the high-tech kit and batteries	~6,100
5 · Commission	18–22	Spin up, pressurise district by district, fill water, seed farms, first residents	—

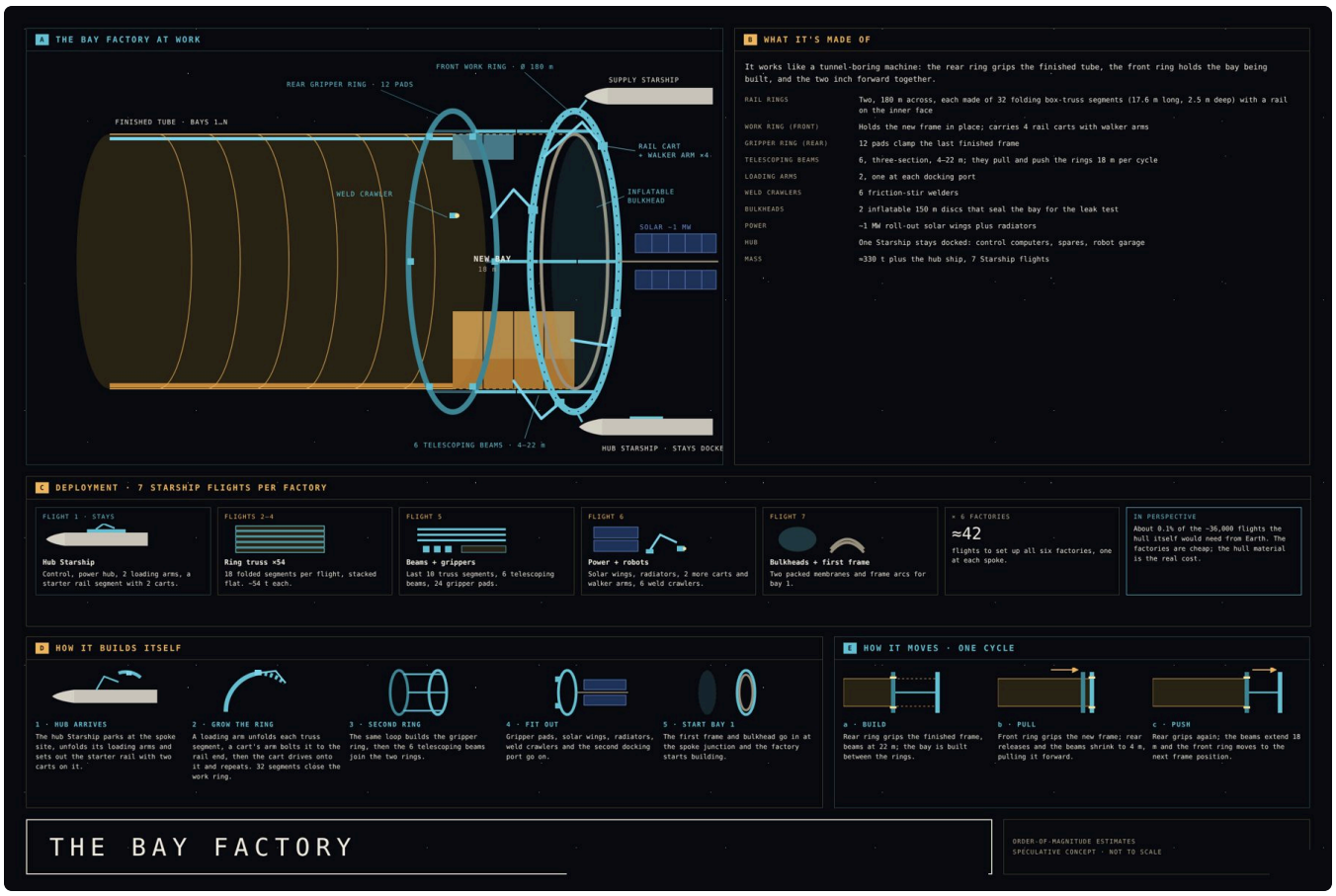
Nitrogen is the swing item. Shipped from Earth, 0.3–0.6 Mt would add 21,000–42,000 launches; from polar ice it adds none.

Build rate. 12 build fronts finishing in 10 years means each front completes about one bay a week:

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\frac{6{,}060 \text{ bays}}{12 \text{ fronts} \times 10 \text{ years}}
\approx 50 \text{ bays per front per year}
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Lunar throughput. ~300 Mt in 10 years is ~30 Mt a year, about 1 tonne every second. At 0.8 kWh per kg, the mass drivers need about 2.7 GW of lunar solar power running continuously.

The bay factory. Two 180 m rail rings and six telescoping beams, deployed in 7 Starship flights.



Deploying and running the bay factory (66 s animation).



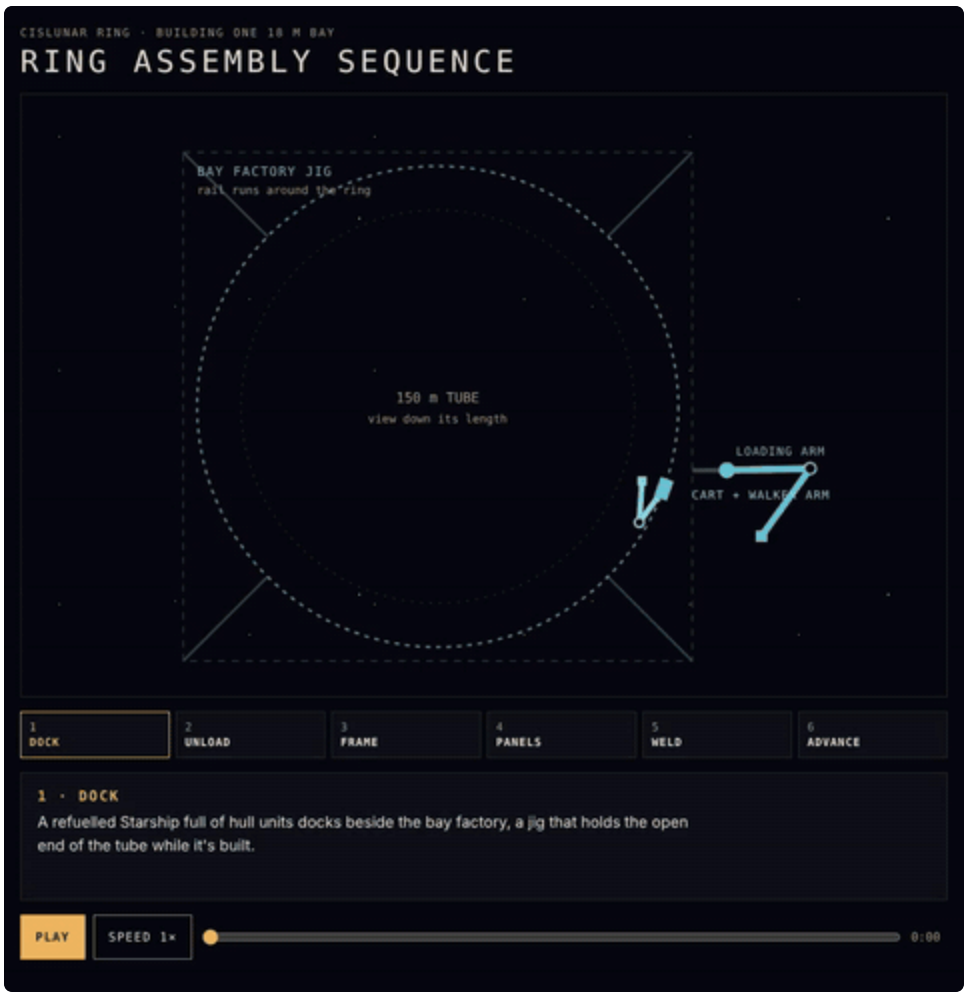
Full video:

Image — The bay factory, full video

Hull units. 16 curved aluminium panels fit in one Starship; 84 panels and one frame make an 18 m bay.



Assembling one bay (57 s animation): dock, unload, close the frame, hang panels, weld and leak-test, advance.



Full video:

Image — Ring assembly sequence, full video

Building the spokes, wings and radiators

The hub goes up first. Six spokes then grow out from it in three opposite pairs, so the structure stays balanced, and the solar wings and radiator fins go on before spin-up, while everything is still in free fall.

Part	How it's built	Size
Spokes	A crawler per spoke walks out on the truss it lays, one 30 m bay at a time, fed by cargo pods running along the finished truss	6 × 17.35 km, ~35 kt each, with an elevator shaft inside
Junction nodes	Each crawler leaves a node at the rim: the first cell of the ring and the start of its bay factories	6
Middle ring	12 small bay factories start at the spokes and build both ways until the arcs meet	r 8.7 km, Ø 40 m, 54.5 km around, ~0.5 Mt
Solar wings	One wing robot per spoke clips ribs to the truss and unrolls two 90 m blankets of lunar-made cells	2.9 km² per spoke, 17 km² in all
Radiator fins	Carriers on the shield rail dock a folded fin into each cell's slot, unfold it, connect the ammonia loops and leak-test it	~650 fins of 150 × 75 m

Why the spokes are fibre. Spinning, a spoke carries its own weight in tension. For a uniform spoke that peaks at the hub:

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\sigma_{\text{root}} = \tfrac{1}{2}\rho\,\omega^2 R^2 = \tfrac{1}{2} \times 2700 \times 5.65\times10^{-4} \times (17{,}350)^2 \approx 230\ \text{MPa}
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That is too close to aluminium's limit, so the load-bearing chords are basalt fibre (over 1 GPa), the same material as the hoop bands.

The middle ring. At half the radius it spins at the same rate but feels **0.5 g**, and its rim moves at 206 m/s, so aluminium carries its own spin at ~115 MPa with no fibre bands. It holds half-gravity industry and labs, the elevator transfer stop, and ties the six spokes together. Placed at 6.6 km instead, it would give Mars gravity (0.38 g) for Mars-preparation research.

Building the spokes and middle ring (48 s animation):

Image — Building the spokes and middle ring

Installing the solar wings and radiator fins (50 s animation):

Image — Installing the solar wings and radiator fins

Ring stacks: several rings on one spine

Instead of flying each counter-rotating pair on its own, six rings (three pairs) can share one non-rotating **spine** 14 km long. Build cost per resident falls with every ring, from **$168–661k for ring 1 to $87–502k for ring 100**. Almost all of that comes from reusing the construction fleet and Earth-made parts getting cheaper; the spine itself is roughly cost-neutral, and what it buys is simpler operations.

How it fits together.

Part	What it is	Key numbers
Spine (centre node)	Non-rotating truss along the shared spin axis, pointed at the Sun, with ports and docks at its head	14 km long, ~200 m across, ~1 Mt
Centre pods	One per ring: a drum on two magnetic bearings round the spine, a motor ring, six spoke sockets and a lock	Ø 400 m, 500 m long, 2 km apart
Ring pairs	Neighbouring rings spin in opposite directions, so each pair has no net spin	3 pairs, ~3.3 million people
Middle rings	One per ring, at half radius	0.5 g, ~0.5 Mt, ~$1.5–5B per ring
Tug Starships	2–4 docked at the spine tips during the build	Steer, trim and damp

Loads on the spine. Tracking the Sun turns the axis about 1° a day, and each ring resists that turn:

$$L = I\omega = 176\ \text{Mt} \times (17.35\ \text{km})^2 \times 0.0238\ \text{s}^{-1} \approx 1.3\times10^{18}\ \text{N·m·s} \qquad \tau = L\,\Omega_{\text{Sun}} \approx 2.5\times10^{11}\ \text{N·m}$$

Neighbouring rings need equal and opposite torques, so the spine carries this between them as bending: about 4 cm of equivalent wall on a 200 m truss (~68 t per metre), and ~0.5 GN on each pod's bearings with them 500 m apart. Spin-up at 2.5 GW pushes ~1.1 × 10^{11} N·m through the pods for about 70 days per pair; in torsion the spine needs under 1 cm of equivalent wall.

Steering with Starships. While the rings aren't spinning, only thrusters can steer the stack, so tug Starships docked at the spine tips do it:

Phase	Attitude	What steers	Propellant
Spine and pods	Axis along the orbit's normal: no gravity-gradient torque, and a stable attitude for ring-shaped bodies	2–4 tug Starships	Small: trims and damping
Rings going up	Same; each ring is built from 12 symmetric fronts so the stack stays balanced	Tugs hold it within ±0.1°	~12 t a day per ring-sized mass at ±0.1°; ~120 t a day at ±1°
First pair spun up	Slews 90° to face the Sun over ~30 days	The spinning pair, each ring tilted ~0.004° against its twin	None
Operating	Sun-pointed; gravity-gradient torque cycles twice an orbit	The pairs, tilting ~0.01°	None

Tilting two counter-rotating rings in opposite directions makes them a scissored gyroscope, storing momentum h with a tilt of only:

$$\delta = \frac{h}{2L} = \frac{\tau_{\text{gg}}/n}{2L} = \frac{4.8\times10^{14}}{2 \times 1.26\times10^{18}} \approx 1.9\times10^{-4}\ \text{rad} \approx 0.011^\circ$$

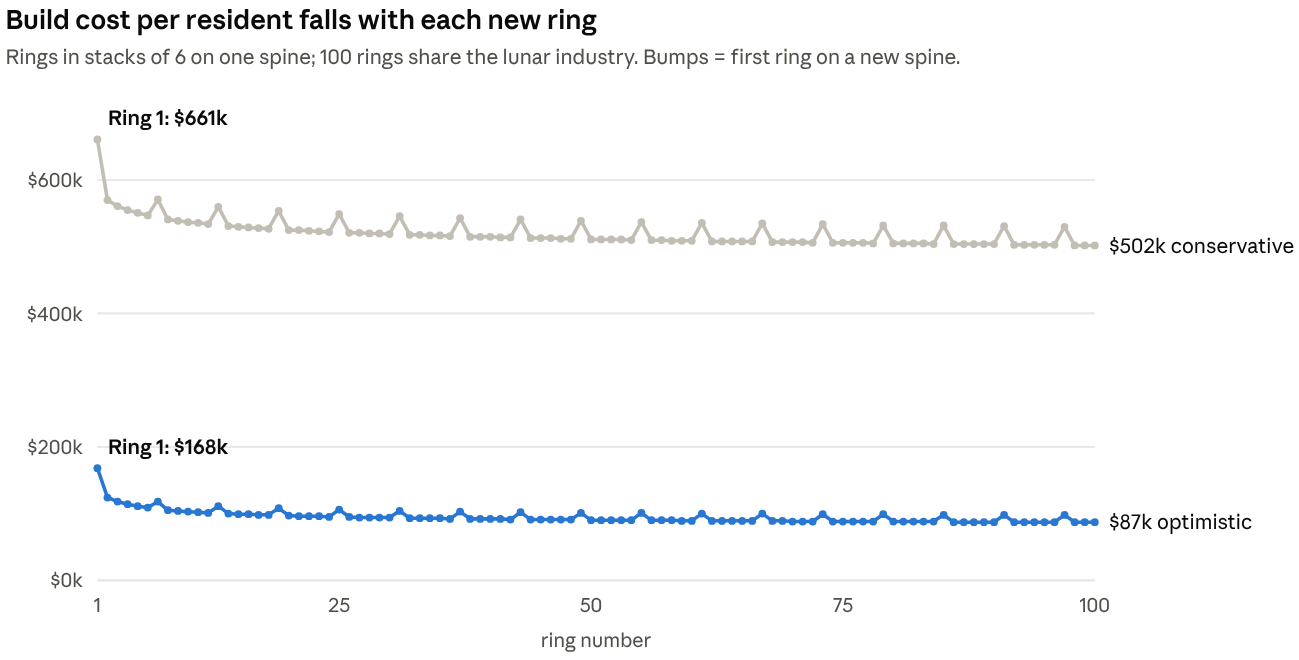
Four tugs carry ~1.8 × 10¹⁰ N·s of impulse between refuellings: 18 m/s for the bare spine, but only 6 cm/s for one finished ring. They steer and trim the structure while it is light; they cannot move a finished stack. Holding a gigatonne stack in its orbit is the open problem, and it argues for a naturally stable orbit.

A stable orbit. The stack circles the Moon backwards about 30,000 km out, once every 4.7 days, while the Moon carries it around Earth. In a simplified Earth–Moon–Sun model it stayed bound for 20 years with no corrections, and adding rings doesn't change the path.

Image — One ring circling the Moon

Shading. The solar wings cover ~1.8% of a ring's disc. Because neighbouring rings turn in opposite directions, their wings sweep past each other, so each ring upstream shades about 2% of the next ring's wings on average; the last ring of six needs ~10% more wing. The tubes line up behind each other too, but they don't use sunlight.

Price per resident for each new ring.



cost model in this section · rings 1–100

Each bump is the first ring on a new spine, which pays for the spine head. Cost per ring and per resident (550,000 people a ring; home price for 2.5 people, using the same 60% share and 40% financing as above):

Ring	Optimistic cost	Per resident	Home	Conservative cost	Per resident	Home
1	$92B	$168k	$0.35M	$364B	$661k	$1.39M
2	$68B	$124k	$0.26M	$314B	$570k	$1.20M
3	$65B	$118k	$0.25M	$309B	$561k	$1.18M
6	$60B	$109k	$0.23M	$301B	$547k	$1.15M
7 (new spine)	$65B	$118k	$0.25M	$314B	$571k	$1.20M
12	$56B	$101k	$0.21M	$294B	$534k	$1.12M
50	$50B	$90k	$0.19M	$281B	$511k	$1.07M
100	$48B	$87k	$0.18M	$276B	$502k	$1.05M

What the model assumes.

- Earth-made kit and power systems get cheaper with experience: 20% (optimistic) or 10% (conservative) per doubling of rings built.
- Bay factories, crawlers and robots move on to the next ring; later rings pay 30% of their cost in refits.
- Design and integration: later rings pay 25%.
- Nitrogen stays the same per ring, and the lunar-industry share stays $5–19B per ring (100 rings).
- Each spine head (docks, gas reserves, control centre, tugs) costs $6–15B and serves six rings; each ring adds a $0.5–1.5B spine segment, replacing $3–9B of its own port, docks, reserves and pointing gear.

Across 100 rings, learning and reuse cut the bill from $8.9T to $5.2T (optimistic) or from $35.6T to $28.7T (conservative). The spine moves either total by about 1%; bigger stacks would save a little more.

If fewer rings are built. If the lunar industry is shared only by the rings actually built, the average cost per resident is far higher at first:

Rings built	Optimistic, per resident	Conservative, per resident
1	$1.07M	$4.08M
2	$592k	$2.31M
6	$266k	$1.12M

Rings built	Optimistic, per resident	Conservative, per resident
12	$181k	$812k
25	$134k	$649k
50	$110k	$567k
100	$95k	$522k

Building the centre pod and the spine (52 s animation):

Image — Building the centre pod and the spine

Building a ring stack (64 s animation):

Image — Building a ring stack

Bootstrapping the systems

Machines keep the first residents alive; living systems take over over 3–5 years. Each stage only starts once the one before it is stable.

Machines keep the first residents alive; living systems take over later



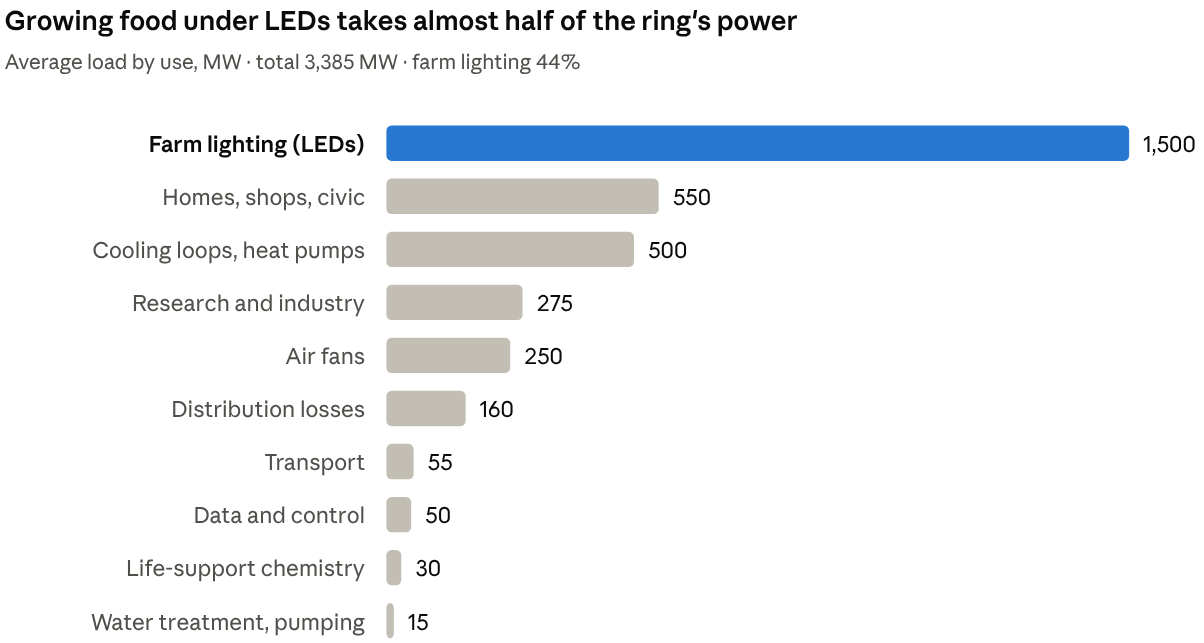
bootstrapping order · 9 stages

1. **Power and cooling (years 8–12).** Solar wings go up on the first spoke with the hub, then the rest as the spokes are built. Radiators and coolant loops go in with them, because every later system makes heat.
2. **Gas and water stockpiles (years 8–18).** Lunar plants stockpile ~0.35 Mt of oxygen and ~0.57 Mt of nitrogen as liquids, and ~5 Mt of water from polar ice, at the hub.
3. **Seal and spin (years 18–19).** The ring can only spin once it is closed. Its ~176 Mt of spinning mass at 412 m/s holds ~4 TWh of rotational energy, about two months of the wings' output, so spin-up takes 2–6 months. Spin needs something to push against, so each ring is built as a counter-rotating pair with a twin (the six cislunar rings are three pairs), driven by motors between them.
4. **Pressurise district by district (years 19–20).** Each of the 606 cells is sealed by pressure doors, so districts fill with air one at a time and are leak-tested before anyone enters.
5. **Machines first (year 20).** District plants run electrolysis, CO_2 scrubbers and reverse-osmosis water recovery, enough for crews and early residents with no help from plants.
6. **Hydroponic farms (years 20–21).** Crops on the farm tiers need no soil, so they start at once and begin making oxygen and food within weeks.
7. **Soil (years 20–23).** Sieved regolith is mixed with compost, digestate and carbon from asteroid material, then inoculated with microbes from Earth. It takes 2–3 years to become living soil.

8. **Forests, wetlands, mountains (years 21–25).** Wetlands and filter beds need months to grow their biofilms; trees need years. Until then, the district plants clean the water and the mountains just store it.
9. **People in waves.** Construction crews of a few thousand, then operators, then residents as each district is certified, reaching ~550,000 around year 25.

Power

The ring uses about **3.4 GW on average** (≈6 kW per person), and growing food under LEDs takes almost half of it.



estimated average loads · 550,000 people

Piping more real sunlight into the farm tiers is the biggest lever: halving the LED share would bring the ring down to about 2.6 GW.

Solar wings. The pair's shared axis points at the Sun, so wings mounted flat along each spoke face it all the time as they turn, with no tracking and no slip rings. The axis swings about 1° a day to follow the Sun through the year; because the twin rings' spins cancel, turning them takes little effort.

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A = \frac{P_{\text{peak}} + P_{\text{recharge}}}{1361\ \text{W/m}^2 \times
0.30 \times 0.85} \times 1.1_{\text{ageing}} \times 1.1_{\text{margin}} =
\frac{4.9\ \text{GW}}{347\ \text{W/m}^2} \times 1.21 \approx 17\ \text{km}^2
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That is ~2.9 km² per spoke: two wings about 90 m wide along ~16 km of spoke, each spoke giving ~1 GW. Made from lunar silicon at ~20% efficiency, the total grows to ~26 km².

Batteries for Earth's shadow. In orbit ~200,000 km out, the ring passes through Earth's shadow (~10,900 km wide there) at 1.4 km/s, so an eclipse lasts up to ~2.1 hours. Eclipses come only in two seasons a year. The internal night is made by louvres, so it needs no stored power.

```
E_{\text{installed}} = \frac{1.6\ \text{GW} \times 2.5\ \text{h}}
{0.8_{\text{depth}}} \times 1.2_{\text{ageing}} = 6\ \text{GWh} \approx
37{,}500\ \text{t of LFP cells}
```

During an eclipse the grid sheds load from 3.4 GW to about 1.6 GW:

Load	Normal	Eclipse
Farm LEDs	1,500 MW	off (plants tolerate a few dark hours)
Research and industry	275 MW	~100 MW, non-critical work paused
Cooling	500 MW	~300 MW, no sunlight coming in
Homes, air, water, life support, data, transport	~960 MW	~960 MW, kept on
Losses	160 MW	~80 MW

Grid. Each spoke feeds a DC ring main in the base through its junction cell. The 30 battery banks (200 MWh each) sit in the district plants, close to the loads, and can carry the ring for hours if one spoke's wings go offline (−1 GW).

Image — Power and energy board: loads, solar wings, radiators, air fans and batteries

Heat and radiators

Soma has to shed about **6 GW of heat** continuously. At 27 °C that takes about **7.3 km² of radiator panel**, best mounted as one fin per cell on the outside of the tube.

Heat balance. About 3.4 GW of electricity (every watt ends up as heat) plus ~2.6 GW of sunlight through the window strips, averaged over the louvre day. The ~3 m regolith shield insulates the hull almost completely, so heat leaves only through radiators. That is ~117 W per m² of hull, or ~11 kW per person.

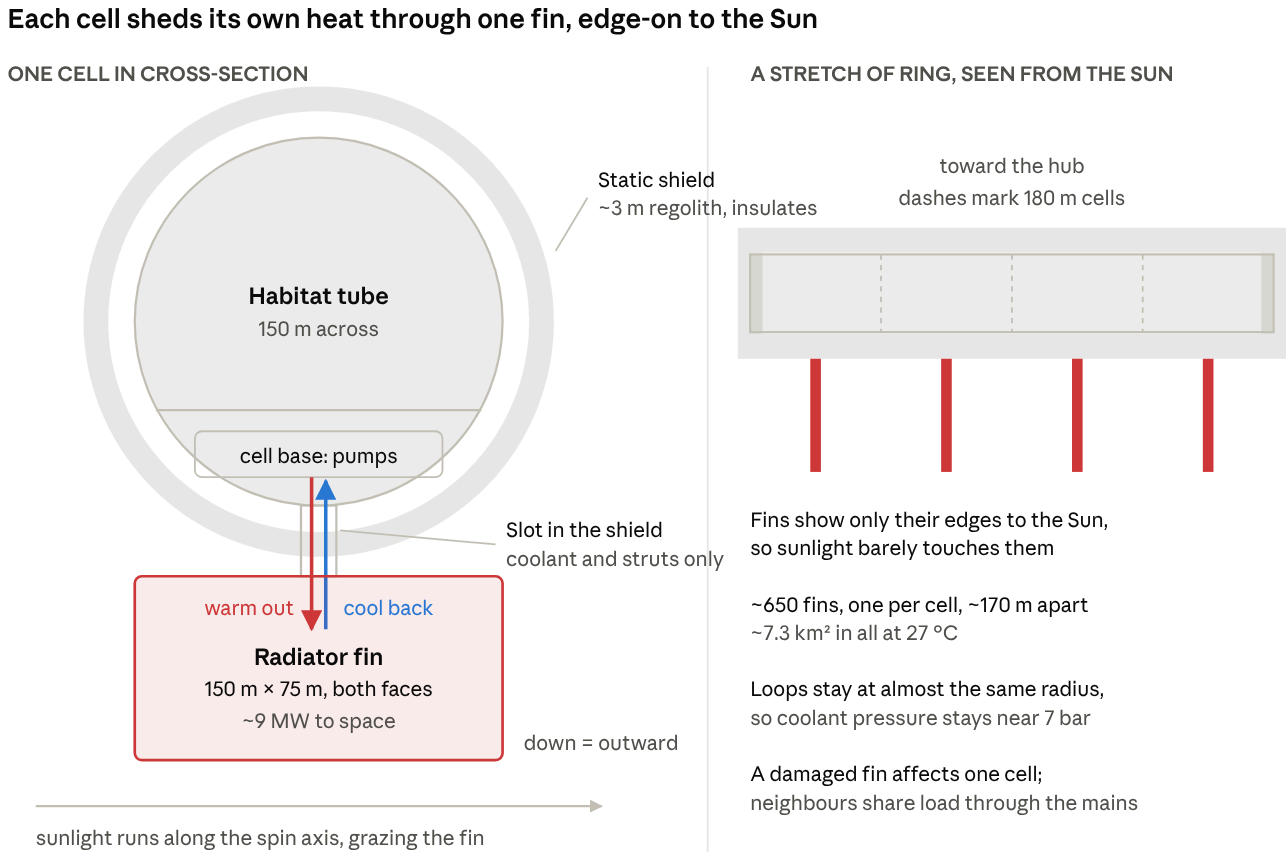
$$q = \varepsilon\, \sigma \left(T_r^{4} - T_{\text{space}}^{4}\right) \times 2_{\text{faces}} = 0.9 \times 5.67\times10^{-8} \times 300^{4} \times 2 \approx 830\ \text{W/m}^2 \qquad A = \frac{6\ \text{GW}}{830\ \text{W/m}^2} \approx 7.3\ \text{km}^2$$

Radiator temperature	Heat per m² of panel (both faces)	Panel area for 6 GW	Notes
7 °C (280 K)	~630 W	~9.6 km²	No heat pumps needed
27 °C (300 K)	~830 W	~7.3 km²	Small heat-pump lift; the baseline
47 °C (320 K)	~1,070 W	~5.6 km²	Heat pumps add ~10–15% more heat
77 °C (350 K)	~1,530 W	~3.9 km²	Smallest, but ~20% more power and heat

Three designs.

Design	Layout	For	Against
A · Spoke fins	Fins along each spoke, ~1.2 km² per spoke	Out of the way; shaded by the solar wings	A 17 km column of coolant builds ~850 bar at the rim; needs staged loops or heat pipes
B · One fin per cell (baseline)	A plate on the tube's outer side, ~75 m deep × 150 m across, through a slot in the shield; ~650 fins	Short loops at nearly constant radius (~7 bar of head); each district cools itself; a failure stays in one cell	~650 shield slots; fins partly see their neighbours, so ~10–15% extra area

Design	Layout	For	Against
C · Radiator skirt	One continuous band around the ring, ~67 m tall	One simple structure	Hard to place without shading the window-strip mirrors; one big shared system



design B · one radiator fin per cell

Each cell's base pumps warm coolant out through a slot in the shield to its own fin and back; the fin's faces point sideways along the ring, so the Sun only ever sees its edge.

Coolant. With a 20 K rise, 6 GW needs ~72 tonnes of coolant a second ring-wide, ~120 kg/s per cell. Water loops run inside the habitat; two-phase ammonia loops (as on the ISS) or heat pipes run in the fins. Warm fluid rises toward the hub in spin gravity, so natural circulation does part of the pumping.

Construction. Aluminium honeycomb panels with embedded heat pipes, ~5–10 kg/m², so ~40–70 kt of lunar aluminium for the ring. White or silvered coatings absorb little sunlight and radiate well. With the spin axis pointed at the Sun, every fin is edge-on to it, and

Earth's glow at 200,000 km is negligible. When the louvres close at night the load drops, and the lakes, river and coolant loops buffer the change.

Air

The air is 57 kPa with 34% oxygen, NASA's exploration atmosphere, and plants supply about 85% of the oxygen; machines in the district plants supply the rest and cover any shortfall.

Gas	Partial pressure	Mass in the ring	Source
Oxygen	~19.4 kPa	~0.35 Mt	Refining lunar regolith
Nitrogen	~37 kPa	~0.57 Mt	Ammonia in polar ice (or Earth)
CO_2	under 0.1 kPa	—	Kept low by plants and scrubbers
Water vapour	~1–1.5 kPa	—	40–60% humidity

Daily balance. People use ~0.84 kg of oxygen and breathe out ~1.04 kg of CO_2 each a day:

$$550{,}000 \times 0.84\ \text{kg} \approx 460\ \text{t O}_2\ \text{a day} \qquad 550{,}000 \times 1.04\ \text{kg} \approx 570\ \text{t CO}_2\ \text{a day}$$

The district plants make the other ~15%, about 70 t of oxygen a day, by electrolysis at ~6 kWh/kg (~18 MW), and a Sabatier reactor turns surplus CO_2 and the hydrogen into methane and water. Methane is cracked into carbon for soil and hydrogen for reuse.

Night-time CO_2. When the louvres close, people and plants both breathe out. Twelve hours of CO_2 (~290 t) in ~0.9 Mt of air raises it by about 300 ppm, which scrubbers handle. LED farm tiers also run on a shifted clock, so some crops are always taking CO_2 in.

Air movement: ~250 MW of fans. Warm air still rises toward the hub in spin gravity, so the fans mostly steer it:

Fan system	What it does	Flow	Power
Cell fan walls	Push air up through each valley or canyon at ~0.1 m/s	~1.1 million m^3/s	~90 MW

Fan system	What it does	Flow	Power
Farm canopy fans	Keep air moving over crops at 0.3–0.5 m/s	—	~120 MW
Return fans and ducts	Draw air from under the window strip back down behind the homes	—	~30 MW
Home ventilation	Heat-recovery units, ~10 L/s per person	~5,500 m³/s	~10 MW

Cell fan power = flow × pressure ÷ efficiency = 1.1 × 10^6 m³/s × 50 Pa ÷ 0.6 ≈ 90 MW. At 57 kPa, air carries ~40% less heat per cubic metre than on Earth, so heat leaves by liquid coolant, not by blowing air.

Losses and top-up. Assuming the welded hull leaks ~0.1% of its air a year, ~900 t a year of nitrogen and oxygen comes from lunar stockpiles.

Water

Over 98% of water is reused. Used water from homes is pumped to the nearest wild cell, lifted into a mountain, and cleaned as spin gravity pulls it down through wetlands and filter layers.

Flow	Amount a day	Path
Household used water	~16,500 m³ (~30 L per person)	Homes → district plant → mountain → river → clean main
Crop transpiration	~100,000 m³	Air-handler condensers → UV → clean main
Drinking and food water	~1,700 m³	Clean main → final RO and UV → homes
Top-up from polar ice	small, replaces leaks and water locked in new soil and biomass	Hub stores → clean main

The mountain loop, step by step.

1. Used water leaves the homes through the used-water main in the base.
2. The district plant's digesters take out the solids and sewage sludge first.
3. A pump in the mountain's hollow core lifts the water ~50–60 m to a summit spring.

4. Reed wetlands on the upper slopes strip nitrogen and phosphorus.
5. Streams and waterfalls carry it down; it soaks into ~2 m of soil, ~1 m of slow sand and ~0.5 m of gravel and biochar, where biofilms clean it over 2–4 days.
6. Drains under the layers collect it into a clean tank inside the mountain, which feeds the river and the clean main.
7. Drinking water gets a final reverse-osmosis and UV pass in each district.

Gravity and pumps. Spin gravity makes "down" point outward, toward the hull, so water runs downhill through every slope and filter layer for free. The ring is level all the way round, though, so nothing flows along it on its own: mains pumps at every cell boundary move water between cells, and weirs with low-lift pumps keep the river moving.

$$P_{\text{lift}} = \frac{\rho\, Q\, g\, h}{\eta} = \frac{1000 \times 0.19\ \text{m}^3/\text{s} \times 9.81 \times 60\ \text{m}}{0.7} \approx 0.16\ \text{MW}$$

Lifting is cheap; moving water along the mains, running the river and treating it brings water to about 15 MW in all.

Enough filter area. 90 wild cells × 180 m × ~130 m of slope is ~2.1 km², about 3.8 m² per person; reed-bed polishing needs roughly 1–5 m² per person.

Image — Landscape and water board: a wild cell's mountains and the district water loop

Food and waste

About 24 km² of crops, ~45 m² per person, feed the ring, and nearly everything that leaves a plate or a body goes back to the farms as fertiliser or soil.

Food. 120 farm cells of nine tiers grow greens, roots, beans and grains under LEDs and light pipes; deep-soil top tiers grow orchards and staples under daylight. Precision fermentation makes proteins and fats, and fish and shrimp live in aquaponics tied into the water loop. A seed bank, crop rotation and spare tiers hold six months of food in reserve. The ring eats about 340 t of food (dry mass) a day.

Waste, per day, ring-wide.

Stream	Amount	Where it goes
Inedible plant matter	~340 t dry	Compost → soil
Food scraps	~55 t	Digesters → biogas + digestate
Sewage solids	~70 t wet	Digesters → biogas + digestate
Urine	~770 m^3	Struvite (phosphorus) and ammonium (nitrogen) → fertiliser; water → mountain loop
Plastics, metals, glass	~110 t	Sorted, melted, reprinted in industry cells
Everything else	small	Plasma gasifier → syngas + glassy slag added to the shield

Nutrient loop. Urine alone returns ~6 t of nitrogen and ~0.3 t of phosphorus to the farms each day. Biogas from the digesters goes to the air plant, where its CO_2 is reused and its methane is cracked into carbon and hydrogen. Losses of potassium and phosphorus are made up from the Moon's KREEP rocks.

How the systems work together

Power is the one system everything depends on, and plants are the one that ties air, water and food together; the design keeps a machine backup behind every living system.

System	Needs	Gives
Power	Sunlight on the wings; radiators to shed heat	Electricity to every other system
Cooling	Power for pumps and heat pumps	Removes ~6 GW of heat from people, lights, machines and sunlight
Air	Power for fans and electrolysis; oxygen from plants; water to split	Breathable air; CO_2 for crops
Water	Power for pumps and treatment; mountains to filter; spin gravity	Drinking and washing water; irrigation; water to split for oxygen; thermal mass

System	Needs	Gives
Farms and forests	Light, water, CO_2, nutrients	Food, ~85% of the oxygen, ~100,000 m^3 of transpired water a day
Waste	Power for digesters and the gasifier	Fertiliser for farms, CO_2 for air, slag for the shield
Mountains	Pumped used water, gravity	Clean water, storage, the landscape

A day. At 06:00 the louvres open: crops start photosynthesising, oxygen rises and CO_2 falls, and transpiration and cooling load climb toward midday. At 18:00 the louvres close and CO_2 creeps up overnight, while LED tiers on a shifted clock keep some crops working. Power stays steady through it all, because the wings outside never see night.

When something fails. Every cell can be sealed, and each district can lean on its neighbours.

Event	First effect	Automatic response	Back to normal
Earth's shadow	Solar power drops to zero	Batteries carry ~1.6 GW; farm LEDs and heavy industry pause	When the ring leaves the shadow, ⩽ 2.5 h
A spoke's wings fail	−1 GW	Batteries bridge; research and industry shed; LED tiers take turns	Repair robots, days to weeks
Hull leak in a cell	Pressure falls in that cell	Pressure doors close; people go to the base refuge floor	Patch robots, hours to days
Fire	34% oxygen makes it spread fast	Evacuate the cell, close it, flood it with nitrogen	Hours
Crop disease	Less food and oxygen from some farm cells	Quarantine those cells; district plants make extra oxygen; food reserve	Months
Mountain filter upset	Water quality falls	Bypass to district-plant reverse osmosis while biofilms recover	Weeks

Event	First effect	Automatic response	Back to normal
District plant outage	Local air and water processing stops	Neighbouring plants carry it through the shared mains	Days
Solar storm	Radiation spike outside	Static shield absorbs it; power and air unaffected	Hours

Image — Life support board: the loops, daily flows and each loop step by step

The full-length videos are also in the chat and in the project repo.

Running and maintenance

Running the ring costs about **$3.4–10.9 billion a year**, roughly $6,000–20,000 per resident, and robots do most of the maintenance with ~25,000 people overseeing them.

Item	Basis	Optimistic a year	Conservative a year
Spare parts from Earth	~1,250 t a year, 2.5% of the high-tech kit	$1.0B	$2.0B
Delivering those parts	1,250 t at $100–700/kg	$0.13B	$0.9B
Battery renewal	6 GWh every ~12 years: 0.5 GWh, ~3,100 t a year	$0.36B	$2.2B
Solar wing renewal	~4% a year, ~0.7 km², lunar-made	$0.1B	$0.5B
Lunar top-ups	air, water and minerals, ~1,000–5,000 t a year	$0.05B	$0.3B
Operations staff	~25,000 people, ~5% of residents	$1.5B	$4.0B
Robots and software	fleet renewal	$0.3B	$1.0B

Item	Basis	Optimistic a year	Conservative a year
Total		**≈ $3.4B**	**≈ $10.9B**

Per household of 2.5 people that is about $15,000–50,000 a year, the yearly upkeep fee each home would pay.

Maintenance plan.

System	Routine	How often	Done by
Hull and seals	Crawler inspection, leak survey	Continuous, full pass monthly	Hull robots
Hoop bands	Strain sensors; fibre inspection	Continuous; robot survey yearly	Sensors, robots
Solar wings	Swap panels hit by micrometeoroids	Monthly	Spoke robots
Radiators	Coolant leak checks, fin swaps	Monthly	Hull robots
Batteries	Cell health checks; bank renewal	Continuous; every 10–15 years	District crews
Fans and pumps	Bearings, filters, motors	Every 3–12 months	District crews
Mountain filters	Scrape sand, replace biochar, harvest reeds	Every 1–3 years	Landscape crews
Water plant	Replace RO membranes; service digesters	Every 3–5 years; yearly	District crews
Farms	Replace LEDs; rotate crops; clean tiers	Every 5–8 years; continuous	Farm robots, growers
Spin and pointing	Pair motors and bearings; daily 1° axis turn	Continuous; overhaul yearly	Hub crew

Risks, open questions and assumptions

The three things most likely to change these numbers are the cost of lunar industry, the nitrogen supply, and whether biological loops this large stay stable.

Biggest risks.

- **Lunar industry.** Nobody has yet built a mass driver, a regolith refinery or a lunar fibre plant; its cost is the least certain line.
- **Nitrogen.** If the polar ice holds less ammonia than LCROSS suggested, nitrogen from Earth could add up to $200B and 40,000 launches per ring.
- **Biology at scale.** Closed ecosystems this large have never run for years. The machine backups are sized to carry the ring on their own, at a higher power cost.
- **Spin-up and pointing.** Each ring needs a counter-rotating twin; a single ring could not be turned to face the Sun.
- **Hoop stress.** At 1 g and 17 km radius, steel cannot carry the spin; the design depends on basalt or glass fibre bands made on the Moon.
- **Density.** Terrace homes (a top floor for everyone) cut the population from ~550,000 to ~375,000 in a mixed layout, which raises the cost per home.

Open questions.

- [ ] How much ammonia do the lunar polar cold traps really hold?
- [ ] What hull leak rate is achievable for 51 km² of welded panels?
- [ ] How much farm lighting can light pipes replace, cutting the 1.5 GW LED load?
- [ ] Which orbit: closer to Earth for shorter trips, or farther out for fewer eclipses?

Assumptions used throughout.

Assumption	Value
Starship launch cost	$1.5M (SpaceX's target) to $10M
Starship payload	~100 t, 8 m payload diameter
Tankers per delivery	~6 to the ring's orbit, ~10–15 to the lunar surface
Rings sharing the lunar industry	100
Solar cells	30% efficient, 85% packing and heat factor
Batteries	LFP, 0.16 kWh/kg, $100/kWh

Assumption	Value
Per-person needs	0.84 kg O_2, 1.04 kg CO_2, ~30 L household water, ~45 m^2 of crops a day
Household size	2.5 people
Atmosphere leak	~0.1% a year

Sources: SpaceX Starship user's guide (payload envelope and mass); Musk's launch-cost target.